

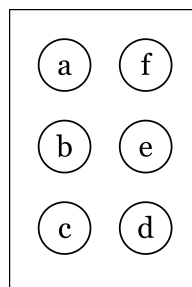
Monotonicity and extrema, group *Alpha* α -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = -8$... ??? ... increasing
 (b) $g(x) = x^2 e^{-4x+5}$... ??? ... $(-\infty, 0)$, $\langle \frac{1}{2}, \infty$
 (c) $h(t) = \frac{\ln t}{3t^4}$??? 2
 (d) $v(x) = 7 - \sqrt[3]{(3x+7)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - 4x + 29}{x - 2}$??? -20
 (f) $\omega(x) = \frac{2}{3}x^3 - \frac{9}{2}x^2 + 9x + 6$, $x \in \langle 1, 5 \rangle$??? 12

1.



Braille letter

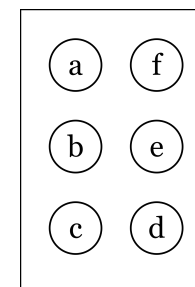
Monotonicity and extrema, group *Alpha* α -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = -11$... ??? ... increasing
 (b) $g(x) = x^2 e^{4x+1}$... ??? ... $(-\infty, 0)$, $\langle -\frac{1}{2}, \infty$
 (c) $h(t) = \frac{\ln t}{3t^3}$??? 1
 (d) $v(x) = 6 - \sqrt[3]{(-7x+2)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - 4x + 19}{x - 3}$??? -16
 (f) $\omega(x) = x^3 - \frac{19}{2}x^2 + 28x - 1$, $x \in \langle 2, 6 \rangle$??? 20

2.



Braille letter

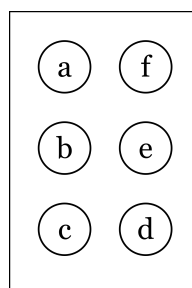
Monotonicity and extrema, group *Alpha* α -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = -5$??? increasing
 (b) $g(x) = x^2 e^{-9x-2}$??? $(-\infty, 0)$, $\langle \frac{2}{9}, \infty$
 (c) $h(t) = \frac{\ln t}{5t^3}$??? 1
 (d) $v(x) = -7 - \sqrt[3]{(-2x-4)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - 3x + 12}{x + 1}$??? 16
 (f) $\omega(x) = \frac{1}{3}x^3 + 5x^2 + 24x - 3$, $x \in \langle -7, -2 \rangle$??? 8

3.



Braille letter

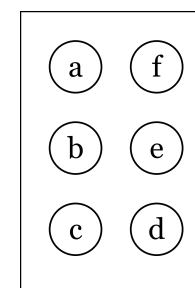
Monotonicity and extrema, group *Alpha* α -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = -12$??? increasing
 (b) $g(x) = x^2 e^{-7x+6}$??? $\langle \frac{2}{7}, 0 \rangle$
 (c) $h(t) = \frac{\ln t}{3t^3}$??? 0
 (d) $v(x) = -9 - \sqrt[3]{(x-3)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 + 9x + 24}{x + 5}$??? -8
 (f) $\omega(x) = 2x^3 - \frac{13}{2}x^2 - 15x + 4$, $x \in \langle -1, 5 \rangle$??? 12

4.



Braille letter

Monotonicity and extrema, group *Beta* β -i

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(a) $f(x) = \frac{x}{9} + \frac{9}{x}$, $x_0 = -11$... ??? ... increasing

(b) $g(x) = x^2 e^{-3x-1}$... ??? ... $(-\infty, 0)$, $\langle \frac{2}{3}, \infty$

(c) $h(t) = \frac{\ln t}{8t^3}$??? 2

(d) $v(x) = -3 - \sqrt[5]{(5x-3)^2}$??? nie

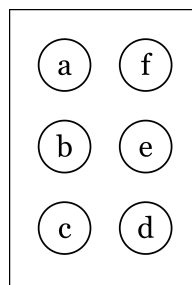
(e) $\varphi(x) = \frac{x^2 + x - 29}{x + 6}$??? -6

(f) $\omega(x) = -\frac{1}{3}x^3 + x^2 + 3x - 3$, $x \in \langle -4, 4 \rangle$??? 4

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

1.



Braille letter

Monotonicity and extrema, group *Beta* β -ii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(a) $f(x) = \frac{x}{8} + \frac{8}{x}$, $x_0 = 1$??? decreasing

(b) $g(x) = x^2 e^{-2x-4}$??? $\langle 1, 0 \rangle$

(c) $h(t) = \frac{\ln t}{t^3}$??? 0

(d) $v(x) = 6 - \sqrt[5]{(-5x-4)^2}$??? nie

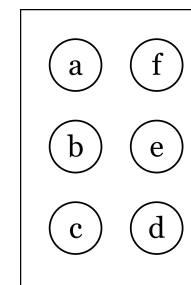
(e) $\varphi(x) = \frac{x^2 - 2x - 7}{x + 2}$??? -4

(f) $\omega(x) = \frac{2}{3}x^3 - \frac{5}{2}x^2 - 3x - 3$, $x \in \langle -1, 5 \rangle$??? 9

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

2.



Braille letter

Monotonicity and extrema, group *Beta* β -iii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = 3$??? decreasing

(b) $g(x) = x^2 e^{x+2}$??? $(-\infty, 0)$, $\langle -2, \infty$

(c) $h(t) = \frac{\ln t}{7t^3}$??? 0

(d) $v(x) = -3 - \sqrt[7]{(8x+2)^2}$??? nie

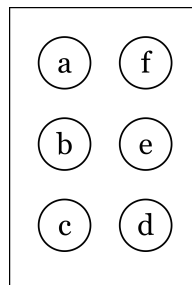
(e) $\varphi(x) = \frac{x^2 - 4x + 5}{x - 2}$??? -4

(f) $\omega(x) = \frac{1}{3}x^3 + \frac{5}{2}x^2 + 4x - 2$, $x \in \langle -5, 1 \rangle$??? 1

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

3.



Braille letter

Monotonicity and extrema, group *Beta* β -iv

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(a) $f(x) = \frac{x}{8} + \frac{8}{x}$, $x_0 = 4$??? increasing

(b) $g(x) = x^2 e^{-6x-4}$... ??? ... $(-\infty, 0)$, $\langle \frac{1}{3}, \infty$

(c) $h(t) = \frac{\ln t}{3t^4}$??? 1

(d) $v(x) = -7 - \sqrt[5]{(-3x-7)^2}$??? nie

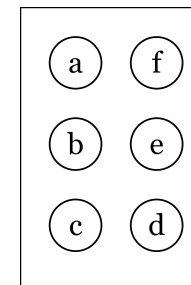
(e) $\varphi(x) = \frac{x^2 + x + 16}{x + 1}$??? -16

(f) $\omega(x) = \frac{1}{3}x^3 + 3x^2 + 8x - 2$, $x \in \langle -5, 0 \rangle$??? 0

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

4.



Braille letter

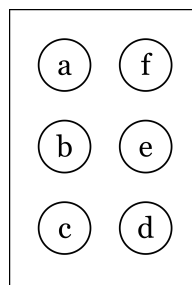
Monotonicity and extrema, group *Gamma* γ -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{8} + \frac{8}{x}$, $x_0 = 5$??? increasing
 (b) $g(x) = x^2 e^{3x-6}$??? $\langle -\frac{2}{3}, 0 \rangle$
 (c) $h(t) = \frac{\ln t}{2t^3}$??? 1
 (d) $v(x) = -6 - \sqrt[3]{(4x+5)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - 3x - 14}{x + 3}$??? 8
 (f) $\omega(x) = -\frac{5}{3}x^3 + 3x^2 + 8x + 1$, $x \in \langle -3, 3 \rangle$??? $\frac{12}{5}$

1.



Braille letter

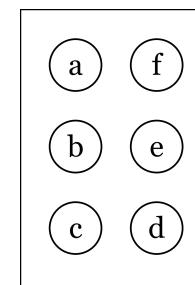
Monotonicity and extrema, group *Gamma* γ -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{8} + \frac{8}{x}$, $x_0 = -11$... ??? ... increasing
 (b) $g(x) = x^2 e^{x+2}$??? $(-\infty, 0)$, $\langle -2, \infty \rangle$
 (c) $h(t) = \frac{\ln t}{3t^5}$??? 1
 (d) $v(x) = 7 - \sqrt[5]{(7x-9)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 + 4x - 1}{x - 1}$??? -8
 (f) $\omega(x) = \frac{4}{3}x^3 - \frac{21}{2}x^2 + 20x + 6$, $x \in \langle 1, 6 \rangle$??? 20

2.



Braille letter

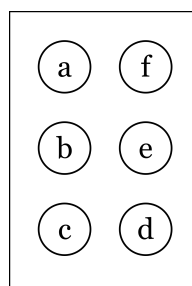
Monotonicity and extrema, group *Gamma* γ -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = -5$... ??? ... increasing
 (b) $g(x) = x^2 e^{-7x-1}$??? $\langle \frac{2}{7}, 0 \rangle$
 (c) $h(t) = \frac{\ln t}{2t^3}$??? 1
 (d) $v(x) = -6 - \sqrt[3]{(-5x+2)^2}$??? áno
 (e) $\varphi(x) = \frac{x^2 - x - 8}{x - 4}$??? 8
 (f) $\omega(x) = \frac{1}{3}x^3 + \frac{1}{2}x^2 - 2x - 2$, $x \in \langle -3, 3 \rangle$??? 1

3.



Braille letter

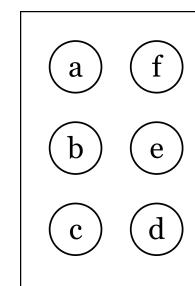
Monotonicity and extrema, group *Gamma* γ -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{5} + \frac{5}{x}$, $x_0 = 3$??? decreasing
 (b) $g(x) = x^2 e^{-4x-4}$... ??? ... $(-\infty, 0)$, $\langle \frac{1}{2}, \infty \rangle$
 (c) $h(t) = \frac{\ln t}{8t^3}$??? 1
 (d) $v(x) = -1 - \sqrt[3]{(3x+2)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - 5x - 5}{x - 6}$??? 4
 (f) $\omega(x) = \frac{4}{3}x^3 + \frac{3}{2}x^2 - 7x - 4$, $x \in \langle -2, 3 \rangle$??? 1

4.



Braille letter

Monotonicity and extrema, group *Delta* δ -i

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = 4$??? decreasing

(b) $g(x) = x^2 e^{3x-1}$??? $\langle -\frac{2}{3}, 0 \rangle$

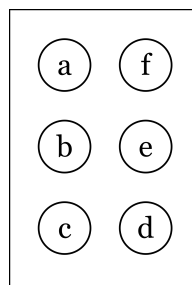
(c) $h(t) = \frac{\ln t}{8t^4}$??? 0

(d) $v(x) = -1 - \sqrt[3]{(5x+7)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 6x - 2}{x - 3}$??? -20

(f) $\omega(x) = -\frac{8}{3}x^3 + \frac{33}{2}x^2 - 4x + 8$, $x \in \langle -2, 5 \rangle$??? $\frac{1}{16}$

1.



Braille letter

Monotonicity and extrema, group *Delta* δ -ii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = -6$??? increasing

(b) $g(x) = x^2 e^{5x+1}$??? $(-\infty, 0)$, $\langle -\frac{2}{5}, \infty \rangle$

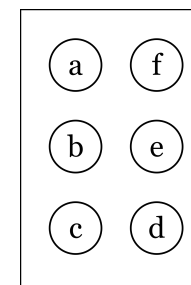
(c) $h(t) = \frac{\ln t}{3t^4}$??? 2

(d) $v(x) = 3 - \sqrt[5]{(x+3)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 6x + 30}{x + 1}$??? 20

(f) $\omega(x) = -\frac{1}{3}x^3 + \frac{9}{2}x^2 - 14x - 1$, $x \in \langle -1, 8 \rangle$??? 10

2.



Braille letter

Monotonicity and extrema, group *Delta* δ -iii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = 1$??? decreasing

(b) $g(x) = x^2 e^{3x-6}$??? $\langle -\frac{2}{3}, 0 \rangle$

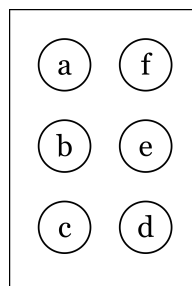
(c) $h(t) = \frac{\ln t}{3t^4}$??? 1

(d) $v(x) = 5 - \sqrt[5]{(-5x+3)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 6x - 23}{x - 3}$??? 8

(f) $\omega(x) = -x^3 + 5x^2 - 3x + 8$, $x \in \langle -2, 4 \rangle$??? $\frac{4}{9}$

3.



Braille letter

Monotonicity and extrema, group *Delta* δ -iv

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = -13$.. ??? .. increasing

(b) $g(x) = x^2 e^{5x+9}$??? $\langle -\frac{2}{5}, 0 \rangle$

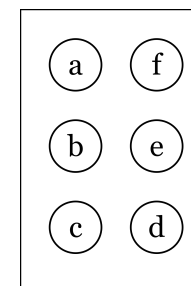
(c) $h(t) = \frac{\ln t}{t^4}$??? 2

(d) $v(x) = 1 - \sqrt[5]{(9x+2)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 3x + 5}{x + 1}$??? -10

(f) $\omega(x) = x^3 - \frac{13}{2}x^2 + 4x + 3$, $x \in \langle 0, 6 \rangle$??? 24

4.



Braille letter

Monotonicity and extrema, group *Epsilon* ϵ -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{5} + \frac{5}{x}$, $x_0 = -9$??? increasing

(b) $g(x) = x^2 e^{6x-1}$??? $\langle -\frac{1}{3}, 0 \rangle$

(c) $h(t) = \frac{\ln t}{8t^5}$??? 1

(d) $v(x) = 2 - \sqrt[3]{(2x-4)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 3x - 17}{x - 3}$??? -4

(f) $\omega(x) = -\frac{2}{3}x^3 + \frac{3}{2}x^2 + 9x + 3$, $x \in \langle -4, 4 \rangle$??? - $\frac{3}{2}$

1.

a

f

b

e

c

d

Braille letter

Monotonicity and extrema, group *Epsilon* ϵ -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = 3$??? decreasing

(b) $g(x) = x^2 e^{4x+5}$??? $(-\infty, 0)$, $\langle -\frac{1}{2}, \infty \rangle$

(c) $h(t) = \frac{\ln t}{5t^3}$??? 2

(d) $v(x) = 4 - \sqrt[7]{(2x-1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 3x - 27}{x + 7}$??? -2

(f) $\omega(x) = -\frac{1}{3}x^3 - \frac{3}{2}x^2 + 4x + 1$, $x \in \langle -7, 2 \rangle$??? 4

2.

a

f

b

e

c

d

Braille letter

Monotonicity and extrema, group *Epsilon* ϵ -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = -6$??? decreasing

(b) $g(x) = x^2 e^{5x-8}$??? $\langle -\frac{2}{5}, 0 \rangle$

(c) $h(t) = \frac{\ln t}{3t^4}$??? 0

(d) $v(x) = 3 - \sqrt[5]{(6x+4)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 3x - 9}{x + 5}$??? 4

(f) $\omega(x) = -\frac{1}{3}x^3 + \frac{3}{2}x^2 + 4x + 3$, $x \in \langle -4, 5 \rangle$??? 4

3.

a

f

b

e

c

d

Braille letter

Monotonicity and extrema, group *Epsilon* ϵ -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = 2$??? decreasing

(b) $g(x) = x^2 e^{-2x-2}$??? $\langle 1, 0 \rangle$

(c) $h(t) = \frac{\ln t}{2t^4}$??? 1

(d) $v(x) = -3 - \sqrt[5]{(3x+1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 7x + 1}{x + 1}$??? -12

(f) $\omega(x) = \frac{1}{3}x^3 + 5x^2 + 24x + 6$, $x \in \langle -7, -2 \rangle$??? 8

4.

a

f

b

e

c

d

Braille letter

Monotonicity and extrema, group *Zeta* ζ -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = 3$??? decreasing

(b) $g(x) = x^2 e^{-9x-2}$... ??? ... $(-\infty, 0)$, $\langle \frac{2}{9}, \infty \rangle$

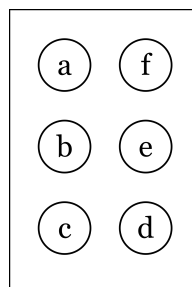
(c) $h(t) = \frac{\ln t}{6t^4}$??? 0

(d) $v(x) = 7 - \sqrt[3]{(6x+2)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 6x + 9}{x - 2}$??? 20

(f) $\omega(x) = \frac{1}{3}x^3 - \frac{1}{2}x^2 - 2x + 2$, $x \in \langle -2, 4 \rangle$??? 4

1.



Braille letter

Monotonicity and extrema, group *Zeta* ζ -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{8} + \frac{8}{x}$, $x_0 = 7$??? increasing

(b) $g(x) = x^2 e^{-5x+6}$??? $(-\infty, 0)$, $\langle \frac{2}{5}, \infty \rangle$

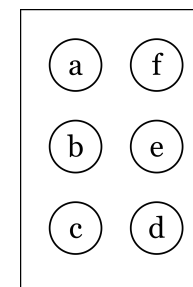
(c) $h(t) = \frac{\ln t}{4t^4}$??? 2

(d) $v(x) = -2 - \sqrt[5]{(9x-5)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + x - 8}{x - 3}$??? -6

(f) $\omega(x) = -\frac{4}{3}x^3 + \frac{21}{2}x^2 - 20x - 3$, $x \in \langle -1, 5 \rangle$??? $-\frac{5}{4}$

2.



Braille letter

Monotonicity and extrema, group *Zeta* ζ -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = -8$??? increasing

(b) $g(x) = x^2 e^{-3x-6}$??? ... $(-\infty, 0)$, $\langle \frac{2}{3}, \infty \rangle$

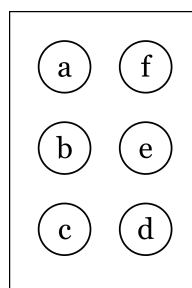
(c) $h(t) = \frac{\ln t}{2t^3}$??? 1

(d) $v(x) = 2 - \sqrt[3]{(2x-3)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 3x - 9}{x + 5}$??? -4

(f) $\omega(x) = -\frac{1}{3}x^3 + x^2 + 3x + 3$, $x \in \langle -4, 4 \rangle$??? -2

3.



Braille letter

Monotonicity and extrema, group *Zeta* ζ -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{12} + \frac{12}{x}$, $x_0 = 9$... ??? ... decreasing

(b) $g(x) = x^2 e^{x+1}$??? ... $(-\infty, 0)$, $\langle -2, \infty \rangle$

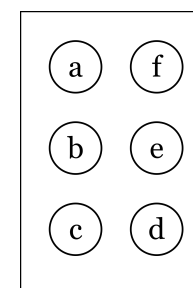
(c) $h(t) = \frac{\ln t}{2t^5}$??? 2

(d) $v(x) = -1 - \sqrt[3]{(9x+4)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + x + 3}{x - 2}$??? -12

(f) $\omega(x) = \frac{1}{3}x^3 + 3x^2 + 8x - 1$, $x \in \langle -5, 0 \rangle$??? 0

4.



Braille letter

Monotonicity and extrema, group *Eta* η -i

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{9} + \frac{9}{x}$, $x_0 = -11$.. ??? .. increasing

(b) $g(x) = x^2 e^{7x-6}$??? $\langle -\frac{2}{7}, 0 \rangle$

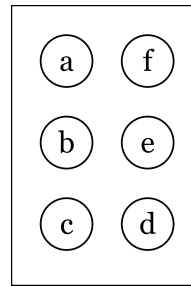
(c) $h(t) = \frac{\ln t}{5t^3}$??? 1

(d) $v(x) = 4 - \sqrt[3]{(-8x-1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 4x + 4}{x - 2}$??? -16

(f) $\omega(x) = \frac{5}{3}x^3 - 8x^2 + 3x + 3$, $x \in \langle 0, 5 \rangle$??? 12

1.



Braille letter

Monotonicity and extrema, group *Eta* η -ii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = 2$??? decreasing

(b) $g(x) = x^2 e^{-3x+7}$??? $\langle \frac{2}{3}, 0 \rangle$

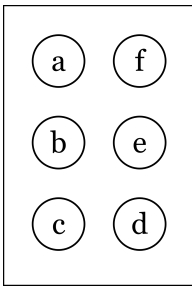
(c) $h(t) = \frac{\ln t}{7t^3}$??? 1

(d) $v(x) = 4 - \sqrt[7]{(4x-2)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 3x + 3}{x - 2}$??? -4

(f) $\omega(x) = \frac{2}{3}x^3 - \frac{1}{2}x^2 - 6x - 2$, $x \in \langle -2, 4 \rangle$??? 4

2.



Braille letter

Monotonicity and extrema, group *Eta* η -iii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{9} + \frac{9}{x}$, $x_0 = 6$??? decreasing

(b) $g(x) = x^2 e^{4x-3}$??? $(-\infty, 0)$, $\langle -\frac{1}{2}, \infty \rangle$

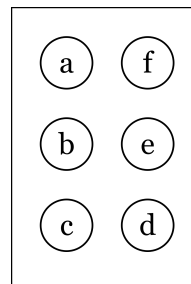
(c) $h(t) = \frac{\ln t}{3t^4}$??? 2

(d) $v(x) = -2 - \sqrt[3]{(4x-1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - x + 16}{x - 1}$??? -18

(f) $\omega(x) = -\frac{1}{3}x^3 - \frac{1}{2}x^2 + 2x - 5$, $x \in \langle -5, 2 \rangle$??? -2

3.



Braille letter

Monotonicity and extrema, group *Eta* η -iv

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{9} + \frac{9}{x}$, $x_0 = 8$??? decreasing

(b) $g(x) = x^2 e^{-5x-6}$??? $\langle \frac{2}{5}, 0 \rangle$

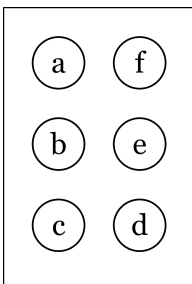
(c) $h(t) = \frac{\ln t}{3t^3}$??? 0

(d) $v(x) = -4 - \sqrt[3]{(-x+7)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 5x + 22}{x - 3}$??? -16

(f) $\omega(x) = \frac{4}{3}x^3 - \frac{13}{2}x^2 + 10x - 7$, $x \in \langle 1, 4 \rangle$??? 8

4.



Braille letter

Monotonicity and extrema, group *Theta* θ -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{2} + \frac{2}{x}$, $x_0 = -3$??? decreasing

(b) $g(x) = x^2 e^{5x-5}$??? $\langle -\frac{2}{5}, 0 \rangle$

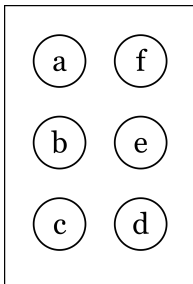
(c) $h(t) = \frac{\ln t}{t^4}$??? 1

(d) $v(x) = -1 - \sqrt[5]{(x+1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 6x + 29}{x - 1}$??? 24

(f) $\omega(x) = \frac{2}{3}x^3 + \frac{9}{2}x^2 + 7x + 2$, $x \in \langle -4, 1 \rangle$??? -1

1.



Braille letter

Monotonicity and extrema, group *Theta* θ -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{2} + \frac{2}{x}$, $x_0 = -9$??? increasing

(b) $g(x) = x^2 e^{2x-1}$??? $(-\infty, 0)$, $\langle -1, \infty \rangle$

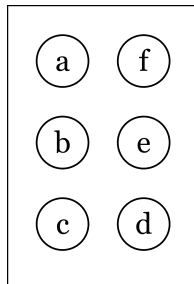
(c) $h(t) = \frac{\ln t}{t^4}$??? 2

(d) $v(x) = 1 - \sqrt[7]{(2x+1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 2x - 11}{x + 3}$??? -6

(f) $\omega(x) = -x^3 + \frac{7}{2}x^2 + 6x - 8$, $x \in \langle -3, 4 \rangle$??? $-\frac{2}{9}$

2.



Braille letter

Monotonicity and extrema, group *Theta* θ -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = -5$??? increasing

(b) $g(x) = x^2 e^{-x+2}$??? $(-\infty, 0)$, $\langle 2, \infty \rangle$

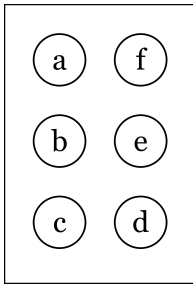
(c) $h(t) = \frac{\ln t}{t^4}$??? 1

(d) $v(x) = 5 - \sqrt[5]{(-8x-2)^2}$??? áno

(e) $\varphi(x) = \frac{x^2 - 4x - 20}{x + 3}$??? -6

(f) $\omega(x) = -\frac{1}{3}x^3 + x^2 + 3x - 5$, $x \in \langle -4, 4 \rangle$??? -2

3.



Braille letter

Monotonicity and extrema, group *Theta* θ -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = -9$... ??? ... increasing

(b) $g(x) = x^2 e^{-5x+6}$??? $\langle \frac{2}{5}, 0 \rangle$

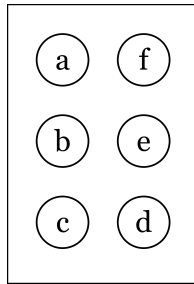
(c) $h(t) = \frac{\ln t}{t^3}$??? 0

(d) $v(x) = -3 - \sqrt[7]{(2x+7)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 2x - 14}{x - 5}$??? -4

(f) $\omega(x) = 2x^3 - \frac{13}{2}x^2 + 2x - 3$, $x \in \langle 0, 4 \rangle$??? 4

4.



Braille letter

Monotonicity and extrema, group *Iota* ι -i

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{5} + \frac{5}{x}$, $x_0 = 4$??? decreasing

(b) $g(x) = x^2 e^{2x-5}$... ??? ... $(-\infty, 0)$, $(-1, \infty)$

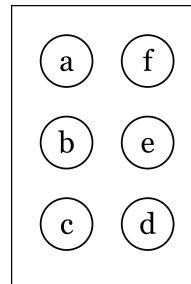
(c) $h(t) = \frac{\ln t}{9t^4}$??? 2

(d) $v(x) = -6 - \sqrt[5]{(-x+1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 5x - 57}{x + 6}$??? -12

(f) $\omega(x) = \frac{4}{3}x^3 - \frac{17}{2}x^2 + 4x - 1$, $x \in \langle 0, 6 \rangle$??? 24

1.



Braille letter

Monotonicity and extrema, group *Iota* ι -ii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = 6$??? decreasing

(b) $g(x) = x^2 e^{-2x+4}$??? $\langle 1, 0 \rangle$

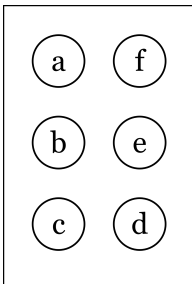
(c) $h(t) = \frac{\ln t}{t^3}$??? 1

(d) $v(x) = -3 - \sqrt[5]{(-3x-4)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 4x + 40}{x - 2}$??? -24

(f) $\omega(x) = -\frac{1}{3}x^3 - \frac{9}{2}x^2 - 14x + 7$, $x \in \langle -10, -1 \rangle$??? -42

2.



Braille letter

Monotonicity and extrema, group *Iota* ι -iii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = -4$??? increasing

(b) $g(x) = x^2 e^{-6x-8}$??? $\langle \frac{1}{3}, 0 \rangle$

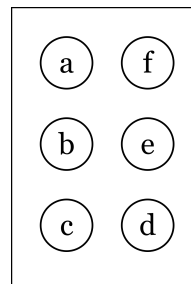
(c) $h(t) = \frac{\ln t}{t^5}$??? 1

(d) $v(x) = -3 - \sqrt[3]{(-3x-3)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 2x + 10}{x + 1}$??? -12

(f) $\omega(x) = \frac{2}{3}x^3 - \frac{19}{2}x^2 + 42x + 3$, $x \in \langle 3, 8 \rangle$??? 36

3.



Braille letter

Monotonicity and extrema, group *Iota* ι -iv

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = 2$??? decreasing

(b) $g(x) = x^2 e^{-2x-2}$??? $(-\infty, 0)$, $\langle 1, \infty \rangle$

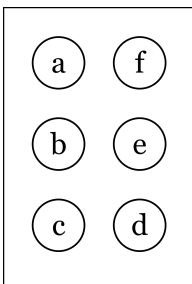
(c) $h(t) = \frac{\ln t}{4t^5}$??? 1

(d) $v(x) = -5 - \sqrt[3]{(-9x-1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 3x - 24}{x + 4}$??? -8

(f) $\omega(x) = -2x^3 + \frac{19}{2}x^2 - 3x - 1$, $x \in \langle -2, 4 \rangle$??? $\frac{1}{9}$

4.



Braille letter

Monotonicity and extrema, group *Kappa* κ -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{9} + \frac{9}{x}$, $x_0 = 7$??? decreasing

(b) $g(x) = x^2 e^{-x+6}$??? $\langle 2, 0 \rangle$

(c) $h(t) = \frac{\ln t}{t^3}$??? 0

(d) $v(x) = -5 - \sqrt[3]{(-x+7)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 5x + 22}{x - 2}$??? 16

(f) $\omega(x) = \frac{2}{3}x^3 - \frac{1}{2}x^2 - 6x - 1$, $x \in \langle -2, 4 \rangle$??? 8

1.

a

f

b

e

c

d

Braille letter

Monotonicity and extrema, group *Kappa* κ -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{2} + \frac{2}{x}$, $x_0 = -3$??? increasing

(b) $g(x) = x^2 e^{x+4}$??? $\langle -2, 0 \rangle$

(c) $h(t) = \frac{\ln t}{2t^3}$??? 1

(d) $v(x) = 2 - \sqrt[3]{(x+8)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 4x - 23}{x - 4}$??? 12

(f) $\omega(x) = -\frac{4}{3}x^3 + \frac{9}{2}x^2 + 9x - 1$, $x \in \langle -3, 4 \rangle$??? 0

2.

a

f

b

e

c

d

Braille letter

Monotonicity and extrema, group *Kappa* κ -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = 6$??? increasing

(b) $g(x) = x^2 e^{4x+1}$??? $\langle -\frac{1}{2}, 0 \rangle$

(c) $h(t) = \frac{\ln t}{5t^3}$??? 2

(d) $v(x) = 8 - \sqrt[3]{(-3x+2)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 9x + 14}{x + 2}$??? 24

(f) $\omega(x) = -\frac{1}{3}x^3 - \frac{11}{2}x^2 - 28x - 2$, $x \in \langle -10, -3 \rangle$??? 70

3.

a

f

b

e

c

d

Braille letter

Monotonicity and extrema, group *Kappa* κ -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{9} + \frac{9}{x}$, $x_0 = 1$??? decreasing

(b) $g(x) = x^2 e^{8x+1}$??? $\langle -\frac{1}{4}, 0 \rangle$

(c) $h(t) = \frac{\ln t}{t^3}$??? 1

(d) $v(x) = -4 - \sqrt[3]{(-7x-5)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - x - 5}{x - 3}$??? -6

(f) $\omega(x) = \frac{5}{3}x^3 + 7x^2 + 9x - 1$, $x \in \langle -2, 1 \rangle$??? -1

4.

a

f

b

e

c

d

Braille letter

Monotonicity and extrema, group *Lambda* λ -i

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = -11$??? increasing

(b) $g(x) = x^2 e^{-2x+4}$??? $(-\infty, 0)$, $\langle 1, \infty$

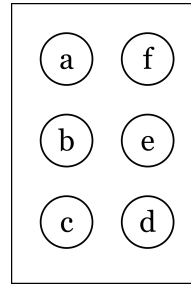
(c) $h(t) = \frac{\ln t}{4t^5}$??? 1

(d) $v(x) = 7 - \sqrt[3]{(-2x-6)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 9x - 48}{x + 4}$??? -6

(f) $\omega(x) = -\frac{1}{3}x^3 + \frac{9}{2}x^2 - 18x - 1$, $x \in \langle 0, 7 \rangle$??? -12

1.



Braille letter

Monotonicity and extrema, group *Lambda* λ -ii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{5} + \frac{5}{x}$, $x_0 = 4$??? decreasing

(b) $g(x) = x^2 e^{-4x-4}$??? $\langle \frac{1}{2}, 0 \rangle$

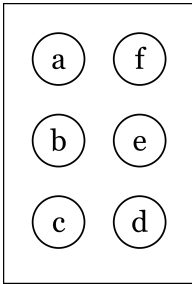
(c) $h(t) = \frac{\ln t}{5t^4}$??? 1

(d) $v(x) = -8 - \sqrt[3]{(8x-5)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 6x + 18}{x + 1}$??? -20

(f) $\omega(x) = \frac{1}{3}x^3 - \frac{1}{2}x^2 - 2x - 2$, $x \in \langle -2, 4 \rangle$??? 4

2.



Braille letter

Monotonicity and extrema, group *Lambda* λ -iii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{5} + \frac{5}{x}$, $x_0 = -6$??? increasing

(b) $g(x) = x^2 e^{-4x+3}$??? $(-\infty, 0)$, $\langle \frac{1}{2}, \infty$

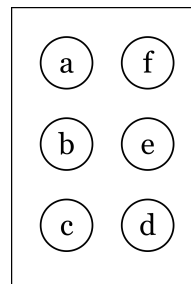
(c) $h(t) = \frac{\ln t}{5t^3}$??? 1

(d) $v(x) = 3 - \sqrt[3]{(3x-7)^2}$??? áno

(e) $\varphi(x) = \frac{x^2 - x + 34}{x - 2}$??? -22

(f) $\omega(x) = -\frac{4}{3}x^3 + \frac{1}{2}x^2 + 14x - 4$, $x \in \langle -4, 3 \rangle$??? $-\frac{21}{8}$

3.



Braille letter

Monotonicity and extrema, group *Lambda* λ -iv

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{2} + \frac{2}{x}$, $x_0 = -4$??? increasing

(b) $g(x) = x^2 e^{-9x-6}$??? $\langle \frac{2}{9}, 0 \rangle$

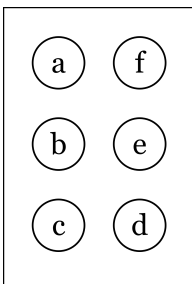
(c) $h(t) = \frac{\ln t}{6t^3}$??? 2

(d) $v(x) = 6 - \sqrt[3]{(-3x-2)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 7x + 19}{x + 2}$??? -12

(f) $\omega(x) = -\frac{7}{3}x^3 + 11x^2 - 3x - 3$, $x \in \langle -2, 4 \rangle$??? $\frac{4}{49}$

4.



Braille letter

Monotonicity and extrema, group $Mu \mu$ -i

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = 1$??? decreasing

(b) $g(x) = x^2 e^{8x-9}$??? $(-\infty, 0)$, $\langle -\frac{1}{4}, \infty$

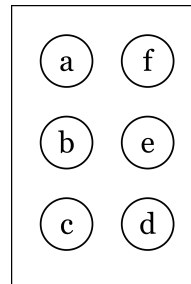
(c) $h(t) = \frac{\ln t}{3t^4}$??? 2

(d) $v(x) = 6 - \sqrt[5]{(8x-5)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 7x + 8}{x-1}$??? -16

(f) $\omega(x) = -\frac{1}{3}x^3 - 3x^2 - 8x - 1$, $x \in \langle -7, -1 \rangle$??? 28

1.



Braille letter

Monotonicity and extrema, group $Mu \mu$ -ii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{5} + \frac{5}{x}$, $x_0 = 4$??? decreasing

(b) $g(x) = x^2 e^{6x+7}$??? $(-\infty, 0)$, $\langle -\frac{1}{3}, \infty$

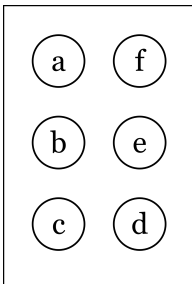
(c) $h(t) = \frac{\ln t}{t^4}$??? 0

(d) $v(x) = -2 - \sqrt[5]{(3x-1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 5x - 13}{x-2}$??? -4

(f) $\omega(x) = -\frac{2}{3}x^3 - \frac{9}{2}x^2 - 7x + 2$, $x \in \langle -6, 0 \rangle$??? 7

2.



Braille letter

Monotonicity and extrema, group $Mu \mu$ -iii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{11} + \frac{11}{x}$, $x_0 = 8$.. ??? .. increasing

(b) $g(x) = x^2 e^{x-3}$??? $\langle -2, 0 \rangle$

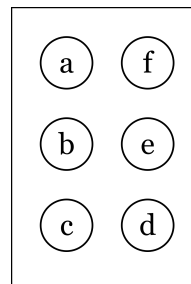
(c) $h(t) = \frac{\ln t}{6t^3}$??? 1

(d) $v(x) = 2 - \sqrt[3]{(8x+1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 3x + 11}{x-2}$??? -10

(f) $\omega(x) = \frac{7}{3}x^3 - 4x^2 + x + 1$, $x \in \langle 0, 3 \rangle$??? 3

3.



Braille letter

Monotonicity and extrema, group $Mu \mu$ -iv

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = 1$??? decreasing

(b) $g(x) = x^2 e^{-2x-1}$??? $\langle 1, 0 \rangle$

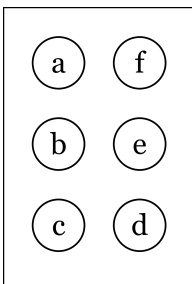
(c) $h(t) = \frac{\ln t}{t^3}$??? 1

(d) $v(x) = -1 - \sqrt[5]{(-4x-4)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 6x + 21}{x+1}$??? 16

(f) $\omega(x) = 2x^3 - \frac{25}{2}x^2 + 4x + 9$, $x \in \langle 0, 6 \rangle$??? 20

4.



Braille letter

Monotonicity and extrema, group $Nu \nu$ -i

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(a) $f(x) = \frac{x}{8} + \frac{8}{x}$, $x_0 = -10$... ??? ... increasing

(b) $g(x) = x^2 e^{-8x-5}$... ??? ... $(-\infty, 0)$, $\langle \frac{1}{4}, \infty$

(c) $h(t) = \frac{\ln t}{3t^5}$??? 0

(d) $v(x) = -2 - \sqrt[3]{(5x+1)^2}$??? nie

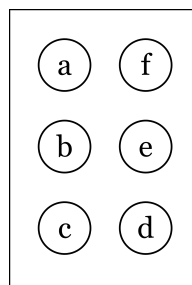
(e) $\varphi(x) = \frac{x^2 - x + 9}{x - 1}$??? -12

(f) $\omega(x) = 2x^3 - \frac{25}{2}x^2 + 4x + 1$, $x \in \langle 0, 6 \rangle$??? 24

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

1.



Braille letter

Monotonicity and extrema, group $Nu \nu$ -ii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(a) $f(x) = \frac{x}{5} + \frac{5}{x}$, $x_0 = 1$??? decreasing

(b) $g(x) = x^2 e^{5x-2}$... ??? ... $(-\infty, 0)$, $\langle -\frac{2}{5}, \infty$

(c) $h(t) = \frac{\ln t}{t^5}$??? 1

(d) $v(x) = 2 - \sqrt[3]{(-3x-2)^2}$??? nie

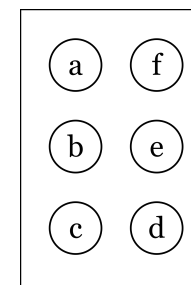
(e) $\varphi(x) = \frac{x^2 - 2x - 31}{x + 5}$??? -8

(f) $\omega(x) = \frac{1}{3}x^3 + \frac{5}{2}x^2 + 4x - 1$, $x \in \langle -5, 1 \rangle$??? 1

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

2.



Braille letter

Monotonicity and extrema, group $Nu \nu$ -iii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(a) $f(x) = \frac{x}{8} + \frac{8}{x}$, $x_0 = -12$??? increasing

(b) $g(x) = x^2 e^{5x+7}$??? $\langle -\frac{2}{5}, 0 \rangle$

(c) $h(t) = \frac{\ln t}{2t^3}$??? 1

(d) $v(x) = -1 - \sqrt[3]{(-2x-1)^2}$??? nie

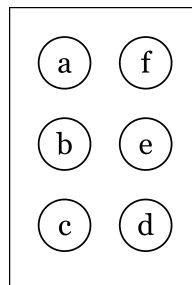
(e) $\varphi(x) = \frac{x^2 + x + 2}{x - 1}$??? 8

(f) $\omega(x) = -2x^3 + \frac{7}{2}x^2 + 10x - 1$, $x \in \langle -3, 3 \rangle$??? $\frac{5}{18}$

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

3.



Braille letter

Monotonicity and extrema, group $Nu \nu$ -iv

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = -5$... ??? ... increasing

(b) $g(x) = x^2 e^{4x-3}$... ??? ... $(-\infty, 0)$, $\langle -\frac{1}{2}, \infty$

(c) $h(t) = \frac{\ln t}{7t^4}$??? 2

(d) $v(x) = -6 - \sqrt[3]{(-x-4)^2}$??? nie

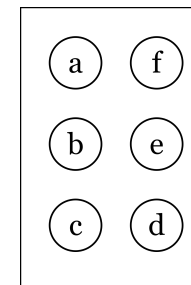
(e) $\varphi(x) = \frac{x^2 - 2x + 9}{x - 2}$??? -12

(f) $\omega(x) = \frac{5}{3}x^3 - \frac{21}{2}x^2 + 4x + 6$, $x \in \langle 0, 6 \rangle$??? 24

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

4.



Braille letter

Monotonicity and extrema, group Xi ξ -i

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(a) $f(x) = \frac{x}{8} + \frac{8}{x}$, $x_0 = 6$??? decreasing

(b) $g(x) = x^2 e^{7x+2}$??? $\langle -\frac{2}{7}, 0 \rangle$

(c) $h(t) = \frac{\ln t}{4t^4}$??? 1

(d) $v(x) = 1 - \sqrt[3]{(-4x+9)^2}$??? nie

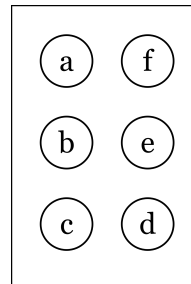
(e) $\varphi(x) = \frac{x^2 + 2x - 11}{x - 3}$??? 8

(f) $\omega(x) = -\frac{1}{3}x^3 - 3x^2 - 5x + 6$, $x \in \langle -8, 0 \rangle$??? 10

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

1.



Braille letter

Monotonicity and extrema, group Xi ξ -ii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = 5$... ??? ... decreasing

(b) $g(x) = x^2 e^{4x-3}$... ??? .. $(-\infty, 0)$, $\langle -\frac{1}{2}, \infty \rangle$

(c) $h(t) = \frac{\ln t}{6t^4}$??? 0

(d) $v(x) = 3 - \sqrt[3]{(-6x-5)^2}$??? nie

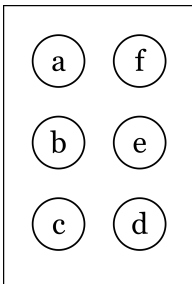
(e) $\varphi(x) = \frac{x^2 + 7x - 2}{x - 2}$??? -16

(f) $\omega(x) = \frac{2}{3}x^3 - \frac{1}{2}x^2 - x - 2$, $x \in \langle -1, 3 \rangle$??? 1

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

2.



Braille letter

Monotonicity and extrema, group Xi ξ -iii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(a) $f(x) = \frac{x}{2} + \frac{2}{x}$, $x_0 = -4$??? increasing

(b) $g(x) = x^2 e^{4x+2}$??? $(-\infty, 0)$, $\langle -\frac{1}{2}, \infty \rangle$

(c) $h(t) = \frac{\ln t}{3t^4}$??? 2

(d) $v(x) = 6 - \sqrt[3]{(-x-7)^2}$??? nie

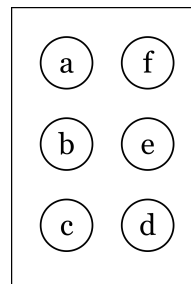
(e) $\varphi(x) = \frac{x^2 - 6x + 4}{x - 6}$??? 8

(f) $\omega(x) = -2x^3 + \frac{19}{2}x^2 - 3x - 7$, $x \in \langle -2, 4 \rangle$??? $-\frac{1}{18}$

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

3.



Braille letter

Monotonicity and extrema, group Xi ξ -iv

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = 4$??? decreasing

(b) $g(x) = x^2 e^{-6x+3}$??? $\langle \frac{1}{3}, 0 \rangle$

(c) $h(t) = \frac{\ln t}{2t^4}$??? 0

(d) $v(x) = -2 - \sqrt[3]{(2x+3)^2}$??? nie

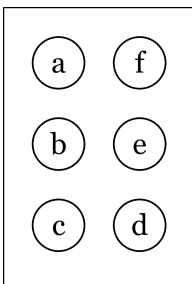
(e) $\varphi(x) = \frac{x^2 - 2x - 59}{x + 7}$??? -8

(f) $\omega(x) = \frac{2}{3}x^3 - \frac{9}{2}x^2 + 4x + 8$, $x \in \langle 0, 6 \rangle$??? 24

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

4.



Braille letter

Monotonicity and extrema, group *Omicron o* -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = 2$... ??? ... increasing

(b) $g(x) = x^2 e^{-4x-1}$.. ??? .. $(-\infty, 0)$, $\langle \frac{1}{2}, \infty \rangle$

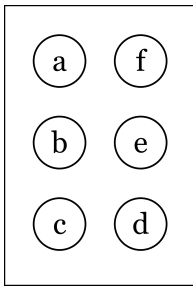
(c) $h(t) = \frac{\ln t}{2t^3}$??? 1

(d) $v(x) = 1 - \sqrt[3]{(4x-6)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 4x + 19}{x + 1}$??? -18

(f) $\omega(x) = \frac{2}{3}x^3 - \frac{1}{2}x^2 - x + 3$, $x \in \langle -1, 3 \rangle$??? 3

1.



Braille letter

Monotonicity and extrema, group *Omicron o* -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{10} + \frac{10}{x}$, $x_0 = 8$... ??? ... decreasing

(b) $g(x) = x^2 e^{-2x-4}$... ??? ... $(-\infty, 0)$, $\langle 1, \infty \rangle$

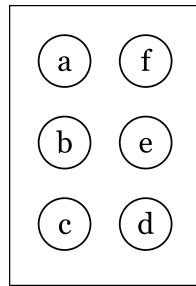
(c) $h(t) = \frac{\ln t}{t^4}$??? 0

(d) $v(x) = -8 - \sqrt[3]{(-5x-6)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 4x - 3}{x - 2}$??? -12

(f) $\omega(x) = \frac{2}{3}x^3 - \frac{9}{2}x^2 + 4x - 6$, $x \in \langle 0, 6 \rangle$??? 20

2.



Braille letter

Monotonicity and extrema, group *Omicron o* -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = -8$... ??? ... increasing

(b) $g(x) = x^2 e^{-4x-9}$??? $\langle \frac{1}{2}, 0 \rangle$

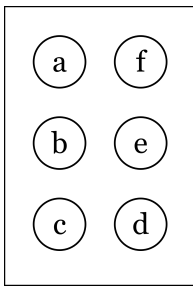
(c) $h(t) = \frac{\ln t}{8t^3}$??? 1

(d) $v(x) = -8 - \sqrt[3]{(4x+3)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 2x + 1}{x + 2}$??? -4

(f) $\omega(x) = \frac{2}{3}x^3 + \frac{3}{2}x^2 - 5x - 3$, $x \in \langle -3, 3 \rangle$??? 1

3.



Braille letter

Monotonicity and extrema, group *Omicron o* -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = -12$??? decreasing

(b) $g(x) = x^2 e^{2x-4}$??? $\langle -1, 0 \rangle$

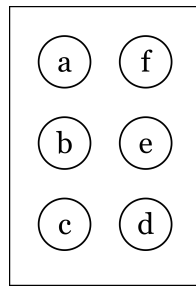
(c) $h(t) = \frac{\ln t}{2t^4}$??? 2

(d) $v(x) = -9 - \sqrt[3]{(3x+4)^2}$??? áno

(e) $\varphi(x) = \frac{x^2 - 6x - 2}{x + 3}$??? -20

(f) $\omega(x) = -\frac{1}{3}x^3 + \frac{9}{2}x^2 - 14x - 7$, $x \in \langle -1, 8 \rangle$??? -2

4.



Braille letter

Monotonicity and extrema, group $Pi \pi$ -i

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = 1$??? decreasing

(b) $g(x) = x^2 e^{3x+1}$??? $\langle -\frac{2}{3}, 0 \rangle$

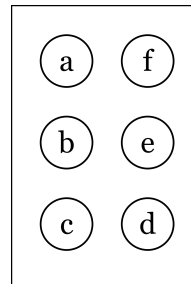
(c) $h(t) = \frac{\ln t}{3t^5}$??? 1

(d) $v(x) = 3 - \sqrt[5]{(-2x+3)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 9x + 15}{x - 7}$??? -4

(f) $\omega(x) = \frac{5}{3}x^3 - \frac{1}{2}x^2 - 18x - 5$, $x \in \langle -2, 4 \rangle$??? 6

1.



Braille letter

Monotonicity and extrema, group $Pi \pi$ -ii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{2} + \frac{2}{x}$, $x_0 = -4$??? increasing

(b) $g(x) = x^2 e^{-7x-9}$??? $\langle \frac{2}{7}, 0 \rangle$

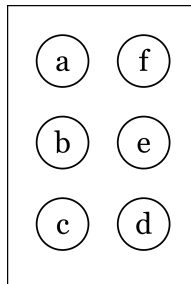
(c) $h(t) = \frac{\ln t}{3t^4}$??? 1

(d) $v(x) = 1 - \sqrt[3]{(-2x-1)^2}$??? áno

(e) $\varphi(x) = \frac{x^2 + x - 52}{x - 7}$??? 8

(f) $\omega(x) = \frac{2}{3}x^3 - \frac{17}{2}x^2 + 30x + 8$, $x \in \langle 2, 8 \rangle$??? 36

2.



Braille letter

Monotonicity and extrema, group $Pi \pi$ -iii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{5} + \frac{5}{x}$, $x_0 = -6$??? increasing

(b) $g(x) = x^2 e^{8x+4}$??? $\langle -\frac{1}{4}, 0 \rangle$

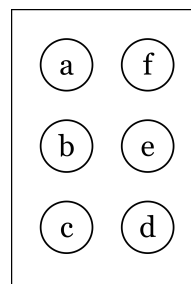
(c) $h(t) = \frac{\ln t}{7t^3}$??? 1

(d) $v(x) = -7 - \sqrt[3]{(3x+1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 5x - 46}{x + 5}$??? -6

(f) $\omega(x) = -\frac{1}{3}x^3 + 4x^2 - 12x + 5$, $x \in \langle -1, 7 \rangle$??? 10

3.



Braille letter

Monotonicity and extrema, group $Pi \pi$ -iv

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = 2$??? decreasing

(b) $g(x) = x^2 e^{7x+5}$... ??? ... $(-\infty, 0)$, $\langle -\frac{2}{7}, \infty \rangle$

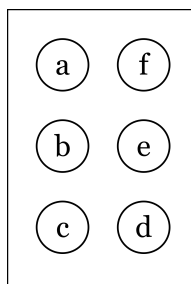
(c) $h(t) = \frac{\ln t}{5t^4}$??? 0

(d) $v(x) = 3 - \sqrt[3]{(-3x+2)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 3x - 24}{x + 5}$??? -16

(f) $\omega(x) = \frac{4}{3}x^3 - \frac{17}{2}x^2 + 15x + 6$, $x \in \langle 1, 5 \rangle$??? 9

4.



Braille letter

Monotonicity and extrema, group *Rho* ρ -i

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = 2$??? increasing

(b) $g(x) = x^2 e^{-5x-6}$... ??? ... $(-\infty, 0)$, $\langle \frac{2}{5}, \infty \rangle$

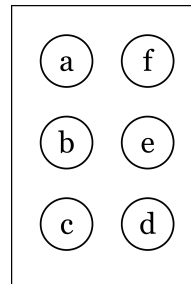
(c) $h(t) = \frac{\ln t}{t^5}$??? 1

(d) $v(x) = 5 - \sqrt[3]{(9x-1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 3x - 53}{x + 6}$??? -4

(f) $\omega(x) = \frac{4}{3}x^3 - \frac{17}{2}x^2 + 4x + 2$, $x \in \langle 0, 6 \rangle$??? 24

1.



Braille letter

Monotonicity and extrema, group *Rho* ρ -ii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = 1$??? decreasing

(b) $g(x) = x^2 e^{8x+2}$??? $\langle -\frac{1}{4}, 0 \rangle$

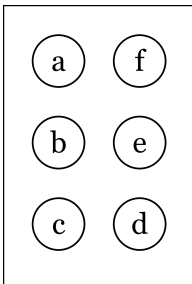
(c) $h(t) = \frac{\ln t}{6t^4}$??? 1

(d) $v(x) = 4 - \sqrt[3]{(-2x-2)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 5x + 40}{x - 1}$??? -24

(f) $\omega(x) = \frac{2}{3}x^3 + \frac{7}{2}x^2 + 5x - 1$, $x \in \langle -3, 1 \rangle$??? 1

2.



Braille letter

Monotonicity and extrema, group *Rho* ρ -iii

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = 5$??? increasing

(b) $g(x) = x^2 e^{-5x-3}$??? $(-\infty, 0)$, $\langle \frac{2}{5}, \infty \rangle$

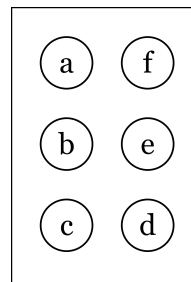
(c) $h(t) = \frac{\ln t}{8t^5}$??? 0

(d) $v(x) = -8 - \sqrt[5]{(-6x-3)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - x - 5}{x + 2}$??? 4

(f) $\omega(x) = -\frac{1}{3}x^3 - 6x^2 - 35x - 1$, $x \in \langle -10, -4 \rangle$??? 70

3.



Braille letter

Monotonicity and extrema, group *Rho* ρ -iv

(a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{5} + \frac{5}{x}$, $x_0 = -9$... ??? ... increasing

(b) $g(x) = x^2 e^{-4x-5}$... ??? ... $(-\infty, 0)$, $\langle \frac{1}{2}, \infty \rangle$

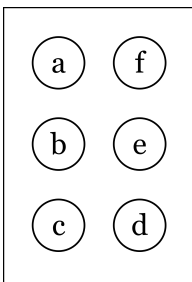
(c) $h(t) = \frac{\ln t}{t^3}$??? 1

(d) $v(x) = -1 - \sqrt[5]{(-7x-2)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 8x + 11}{x + 7}$??? -6

(f) $\omega(x) = x^3 - \frac{7}{2}x^2 - 6x + 6$, $x \in \langle -1, 5 \rangle$??? 15

4.



Braille letter

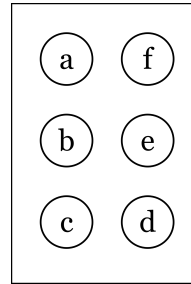
Monotonicity and extrema, group *Sigma* σ -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = -6$??? decreasing
 (b) $g(x) = x^2 e^{5x+1}$??? $\langle -\frac{2}{5}, 0 \rangle$
 (c) $h(t) = \frac{\ln t}{4t^5}$??? 1
 (d) $v(x) = 3 - \sqrt[5]{(-3x-1)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 + 3x - 31}{x - 5}$??? 12
 (f) $\omega(x) = -\frac{1}{3}x^3 + \frac{5}{2}x^2 - 4x + 4$, $x \in \langle -2, 5 \rangle$??? -2

1.



Braille letter

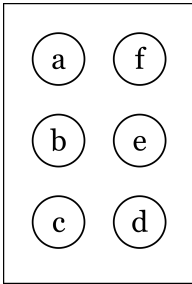
Monotonicity and extrema, group *Sigma* σ -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = 1$??? decreasing
 (b) $g(x) = x^2 e^{-2x-4}$??? $\langle 1, 0 \rangle$
 (c) $h(t) = \frac{\ln t}{t^5}$??? 1
 (d) $v(x) = -4 - \sqrt[3]{(-6x+1)^2}$??? áno
 (e) $\varphi(x) = \frac{x^2 - 3x - 31}{x + 5}$??? -14
 (f) $\omega(x) = \frac{2}{3}x^3 + \frac{7}{2}x^2 + 5x - 2$, $x \in \langle -3, 1 \rangle$??? 1

2.



Braille letter

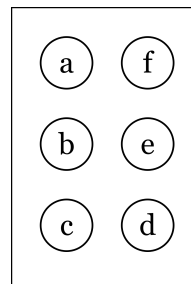
Monotonicity and extrema, group *Sigma* σ -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = -6$... ??? ... decreasing
 (b) $g(x) = x^2 e^{3x-2}$??? $\langle -\frac{2}{3}, 0 \rangle$
 (c) $h(t) = \frac{\ln t}{t^4}$??? 0
 (d) $v(x) = 5 - \sqrt[3]{(x+3)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 + 6x + 9}{x + 2}$??? -6
 (f) $\omega(x) = \frac{4}{3}x^3 - \frac{13}{2}x^2 + 10x - 9$, $x \in \langle 1, 4 \rangle$??? 8

3.



Braille letter

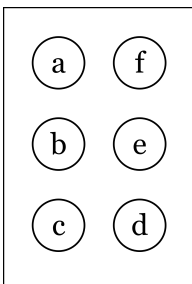
Monotonicity and extrema, group *Sigma* σ -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = 3$??? increasing
 (b) $g(x) = x^2 e^{5x+5}$??? $\langle -\frac{2}{5}, 0 \rangle$
 (c) $h(t) = \frac{\ln t}{t^3}$??? 1
 (d) $v(x) = 5 - \sqrt[5]{(8x-2)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 + 7x - 56}{x - 5}$??? -8
 (f) $\omega(x) = -\frac{2}{3}x^3 + \frac{3}{2}x^2 + 2x + 4$, $x \in \langle -3, 3 \rangle$??? $\frac{3}{2}$

4.



Braille letter

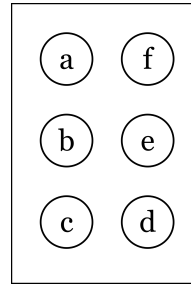
Monotonicity and extrema, group *Tau* τ -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{2} + \frac{2}{x}$, $x_0 = -6$??? increasing
 (b) $g(x) = x^2 e^{-2x+7}$??? $\langle 1, 0 \rangle$
 (c) $h(t) = \frac{\ln t}{t^5}$??? 1
 (d) $v(x) = 1 - \sqrt[3]{(-6x+3)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 + 2x + 1}{x + 2}$??? -4
 (f) $\omega(x) = -\frac{4}{3}x^3 + \frac{13}{2}x^2 - 3x + 1$, $x \in \langle -2, 4 \rangle$??? - $\frac{1}{2}$

1.



Braille letter

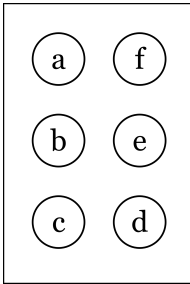
Monotonicity and extrema, group *Tau* τ -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{8} + \frac{8}{x}$, $x_0 = 4$??? decreasing
 (b) $g(x) = x^2 e^{4x+4}$??? $(-\infty, 0)$, $\langle -\frac{1}{2}, \infty \rangle$
 (c) $h(t) = \frac{\ln t}{5t^3}$??? 1
 (d) $v(x) = -1 - \sqrt[5]{(2x-4)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 + 6x - 31}{x - 4}$??? -12
 (f) $\omega(x) = -\frac{1}{3}x^3 - \frac{7}{2}x^2 - 6x - 9$, $x \in \langle -9, 0 \rangle$??? 18

2.



Braille letter

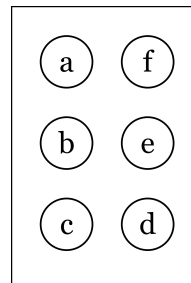
Monotonicity and extrema, group *Tau* τ -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = 6$??? increasing
 (b) $g(x) = x^2 e^{x-7}$??? $\langle -2, 0 \rangle$
 (c) $h(t) = \frac{\ln t}{4t^3}$??? 1
 (d) $v(x) = 2 - \sqrt[5]{(4x+1)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - 3x - 17}{x + 3}$??? -4
 (f) $\omega(x) = -\frac{1}{3}x^3 - 6x^2 - 32x + 1$, $x \in \langle -11, -3 \rangle$??? 88

3.



Braille letter

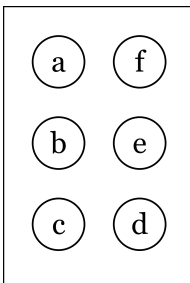
Monotonicity and extrema, group *Tau* τ -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = -9$??? increasing
 (b) $g(x) = x^2 e^{-3x-5}$??? $\langle \frac{2}{3}, 0 \rangle$
 (c) $h(t) = \frac{\ln t}{2t^4}$??? 0
 (d) $v(x) = -3 - \sqrt[5]{(7x+6)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - 4x + 5}{x - 2}$??? -4
 (f) $\omega(x) = -\frac{2}{3}x^3 - \frac{1}{2}x^2 + 10x + 8$, $x \in \langle -5, 3 \rangle$??? $\frac{5}{2}$

4.



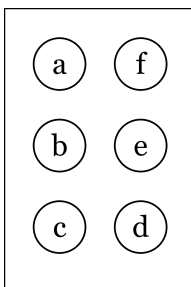
Braille letter

Monotonicity and extrema, group *Upsilon v* -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

1.



Braille letter

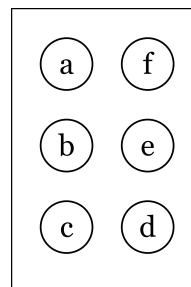
- (a) $f(x) = \frac{x}{8} + \frac{8}{x}$, $x_0 = 7$??? decreasing
 (b) $g(x) = x^2 e^{3x-3}$... ??? ... $(-\infty, 0)$, $\langle -\frac{2}{3}, \infty \rangle$
 (c) $h(t) = \frac{\ln t}{6t^3}$??? 1
 (d) $v(x) = -9 - \sqrt[3]{(-5x+2)^2}$??? áno
 (e) $\varphi(x) = \frac{x^2 - 8x + 40}{x - 3}$??? -20
 (f) $\omega(x) = -x^3 + \frac{7}{2}x^2 + 6x - 2$, $x \in \langle -3, 4 \rangle$??? 2

Monotonicity and extrema, group *Upsilon v* -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

2.



Braille letter

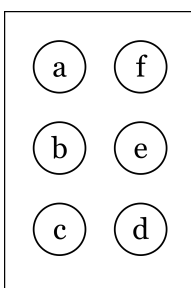
- (a) $f(x) = \frac{x}{5} + \frac{5}{x}$, $x_0 = -9$... ??? ... increasing
 (b) $g(x) = x^2 e^{-x+2}$??? $\langle 2, 0 \rangle$
 (c) $h(t) = \frac{\ln t}{t^5}$??? 0
 (d) $v(x) = 3 - \sqrt[3]{(-2x+8)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - 6x - 15}{x + 2}$??? -4
 (f) $\omega(x) = \frac{1}{3}x^3 - 3x^2 + 8x + 2$, $x \in \langle 1, 6 \rangle$??? 20

Monotonicity and extrema, group *Upsilon v* -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

3.



Braille letter

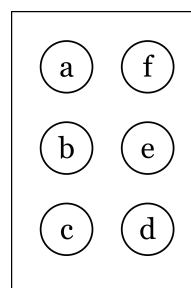
- (a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = -8$??? increasing
 (b) $g(x) = x^2 e^{6x+1}$??? ... $(-\infty, 0)$, $\langle -\frac{1}{3}, \infty \rangle$
 (c) $h(t) = \frac{\ln t}{2t^3}$??? 0
 (d) $v(x) = -1 - \sqrt[3]{(2x-1)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - 3x + 5}{x + 1}$??? 12
 (f) $\omega(x) = -\frac{4}{3}x^3 - \frac{3}{2}x^2 + 7x - 2$, $x \in \langle -4, 2 \rangle$??? $\frac{7}{4}$

Monotonicity and extrema, group *Upsilon v* -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

4.



Braille letter

- (a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = -12$??? increasing
 (b) $g(x) = x^2 e^{-4x-3}$??? ... $(-\infty, 0)$, $\langle \frac{1}{2}, \infty \rangle$
 (c) $h(t) = \frac{\ln t}{t^4}$??? 1
 (d) $v(x) = -5 - \sqrt[3]{(5x-1)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 + 8x + 21}{x + 2}$??? -12
 (f) $\omega(x) = -\frac{2}{3}x^3 - \frac{1}{2}x^2 + 10x + 3$, $x \in \langle -5, 3 \rangle$??? -5

Monotonicity and extrema, group *Phi* ϕ -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = -9$??? increasing

(b) $g(x) = x^2 e^{3x-4}$??? $\langle -\frac{2}{3}, 0 \rangle$

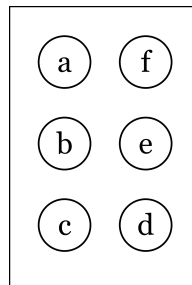
(c) $h(t) = \frac{\ln t}{2t^4}$??? 2

(d) $v(x) = -5 - \sqrt[3]{(x+6)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + x - 8}{x + 4}$??? -8

(f) $\omega(x) = -\frac{4}{3}x^3 + \frac{1}{2}x^2 + 14x - 5$, $x \in \langle -4, 3 \rangle$??? 7

1.



Braille letter

Monotonicity and extrema, group *Phi* ϕ -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{8} + \frac{8}{x}$, $x_0 = 5$??? decreasing

(b) $g(x) = x^2 e^{-8x+3}$??? $\langle \frac{1}{4}, 0 \rangle$

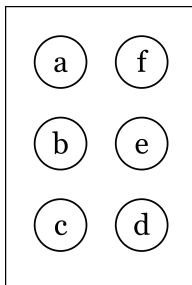
(c) $h(t) = \frac{\ln t}{t^3}$??? 2

(d) $v(x) = 6 - \sqrt[5]{(-2x+1)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 2x + 25}{x + 2}$??? -22

(f) $\omega(x) = -\frac{1}{3}x^3 + \frac{7}{2}x^2 - 10x - 2$, $x \in \langle -1, 6 \rangle$??? 10

2.



Braille letter

Monotonicity and extrema, group *Phi* ϕ -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = 2$??? decreasing

(b) $g(x) = x^2 e^{-2x+6}$??? $\langle 1, 0 \rangle$

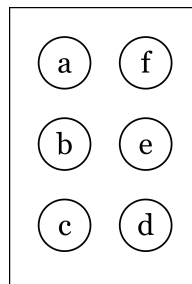
(c) $h(t) = \frac{\ln t}{6t^3}$??? 1

(d) $v(x) = -1 - \sqrt[3]{(-5x+3)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 5x - 27}{x + 4}$??? 12

(f) $\omega(x) = -x^3 + 2x^2 + 4x + 6$, $x \in \langle -3, 3 \rangle$??? 2

3.



Braille letter

Monotonicity and extrema, group *Phi* ϕ -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

(a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = -4$??? increasing

(b) $g(x) = x^2 e^{-x-3}$??? $\langle 2, 0 \rangle$

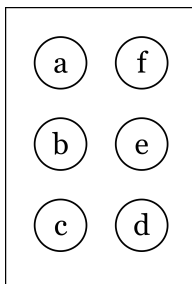
(c) $h(t) = \frac{\ln t}{2t^5}$??? 2

(d) $v(x) = 1 - \sqrt[7]{(3x-3)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 9x + 12}{x - 1}$??? -8

(f) $\omega(x) = -x^3 - x^2 + 5x - 2$, $x \in \langle -4, 2 \rangle$??? $-\frac{20}{9}$

4.



Braille letter

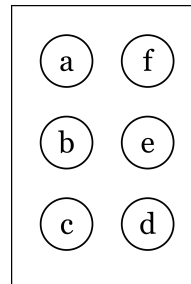
Monotonicity and extrema, group *Chi* χ -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{3} + \frac{3}{x}$, $x_0 = -6$??? decreasing
 (b) $g(x) = x^2 e^{-x+6}$??? $(-\infty, 0)$, $\langle 2, \infty \rangle$
 (c) $h(t) = \frac{\ln t}{3t^5}$??? 1
 (d) $v(x) = -4 - \sqrt[7]{(-4x+8)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - x - 38}{x+6}$??? -8
 (f) $\omega(x) = -x^3 + \frac{1}{2}x^2 + 10x - 7$, $x \in \langle -4, 3 \rangle$??? $\frac{20}{3}$

1.



Braille letter

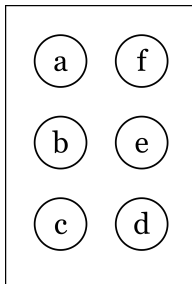
Monotonicity and extrema, group *Chi* χ -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = 6$??? decreasing
 (b) $g(x) = x^2 e^{-2x+4}$... ??? ... $(-\infty, 0)$, $\langle 1, \infty \rangle$
 (c) $h(t) = \frac{\ln t}{2t^5}$??? 1
 (d) $v(x) = -3 - \sqrt[5]{(3x+3)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 + 7x + 35}{x+2}$??? -20
 (f) $\omega(x) = 3x^3 - \frac{37}{2}x^2 + 4x - 5$, $x \in \langle 0, 6 \rangle$??? 20

2.



Braille letter

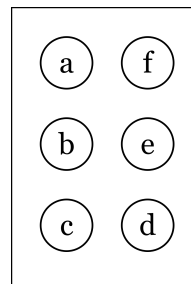
Monotonicity and extrema, group *Chi* χ -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{5} + \frac{5}{x}$, $x_0 = 1$??? decreasing
 (b) $g(x) = x^2 e^{5x+3}$... ??? ... $(-\infty, 0)$, $\langle -\frac{2}{5}, \infty \rangle$
 (c) $h(t) = \frac{\ln t}{3t^3}$??? 0
 (d) $v(x) = 6 - \sqrt[7]{(3x-2)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - 4x + 7}{x-3}$??? -8
 (f) $\omega(x) = \frac{4}{3}x^3 - \frac{17}{2}x^2 + 4x - 8$, $x \in \langle 0, 6 \rangle$??? 20

3.



Braille letter

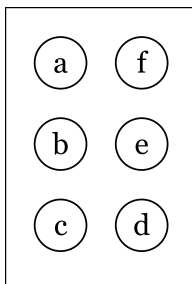
Monotonicity and extrema, group *Chi* χ -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = -9$??? increasing
 (b) $g(x) = x^2 e^{2x+2}$??? $(-\infty, 0)$, $\langle -1, \infty \rangle$
 (c) $h(t) = \frac{\ln t}{t^4}$??? 0
 (d) $v(x) = 5 - \sqrt[7]{(-4x+6)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 + 6x - 26}{x-3}$??? -4
 (f) $\omega(x) = -\frac{2}{3}x^3 + \frac{5}{2}x^2 + 3x + 4$, $x \in \langle -3, 4 \rangle$??? $-\frac{1}{2}$

4.



Braille letter

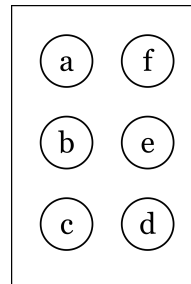
Monotonicity and extrema, group *Psi* ψ -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{7} + \frac{7}{x}$, $x_0 = 6$??? decreasing
 (b) $g(x) = x^2 e^{5x+7}$... ??? ... $(-\infty, 0)$, $\langle -\frac{2}{5}, \infty$
 (c) $h(t) = \frac{\ln t}{3t^4}$??? 1
 (d) $v(x) = -6 - \sqrt[3]{(5x-4)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - 5x + 7}{x - 2}$??? -4
 (f) $\omega(x) = \frac{7}{3}x^3 - \frac{29}{2}x^2 + 4x - 4$, $x \in \langle 0, 6 \rangle$??? 24

1.



Braille letter

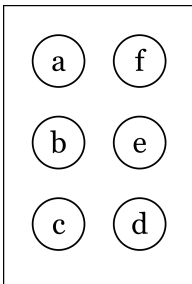
Monotonicity and extrema, group *Psi* ψ -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{5} + \frac{5}{x}$, $x_0 = -6$??? decreasing
 (b) $g(x) = x^2 e^{-2x-5}$??? $(-\infty, 0)$, $\langle 1, \infty$
 (c) $h(t) = \frac{\ln t}{7t^4}$??? 2
 (d) $v(x) = -3 - \sqrt[3]{(-8x+6)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 + x + 10}{x - 2}$??? 16
 (f) $\omega(x) = \frac{2}{3}x^3 + \frac{15}{2}x^2 + 27x + 9$, $x \in \langle -5, -1 \rangle$??? 3

2.



Braille letter

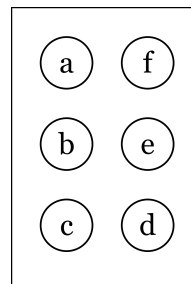
Monotonicity and extrema, group *Psi* ψ -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{9} + \frac{9}{x}$, $x_0 = 8$??? decreasing
 (b) $g(x) = x^2 e^{8x-6}$??? $(-\infty, 0)$, $\langle -\frac{1}{4}, \infty$
 (c) $h(t) = \frac{\ln t}{4t^4}$??? 1
 (d) $v(x) = 2 - \sqrt[3]{(-6x+4)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 - 4x + 4}{x + 1}$??? -12
 (f) $\omega(x) = -\frac{1}{3}x^3 - 4x^2 - 12x + 1$, $x \in \langle -9, -1 \rangle$??? 54

3.



Braille letter

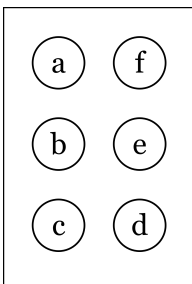
Monotonicity and extrema, group *Psi* ψ -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .
 (b) Determine the intervals on which the function $g(x)$ is decreasing.
 (c) Determine the number of stationary points of the function $h(t)$.
 (d) Does the function $v(x)$ attain a local extremum?

- (e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.
 (f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = -12$.. ??? .. increasing
 (b) $g(x) = x^2 e^{7x-6}$.. ??? .. $(-\infty, 0)$, $\langle -\frac{2}{7}, \infty$
 (c) $h(t) = \frac{\ln t}{8t^4}$??? 2
 (d) $v(x) = -6 - \sqrt[3]{(x+4)^2}$??? nie
 (e) $\varphi(x) = \frac{x^2 + 5x + 1}{x + 5}$??? -4
 (f) $\omega(x) = x^3 + \frac{5}{2}x^2 - 8x + 2$, $x \in \langle -3, 3 \rangle$??? 1

4.



Braille letter

Monotonicity and extrema, group *Omega* ω -i

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{4} + \frac{4}{x}$, $x_0 = -6$... ??? ... decreasing

(b) $g(x) = x^2 e^{x+5}$??? $\langle -2, 0 \rangle$

(c) $h(t) = \frac{\ln t}{t^5}$??? 0

(d) $v(x) = -2 - \sqrt[7]{(6x+7)^2}$??? áno

(e) $\varphi(x) = \frac{x^2 - 6x + 6}{x - 5}$??? -4

(f) $\omega(x) = \frac{8}{3}x^3 - \frac{25}{2}x^2 + 3x - 4$, $x \in \langle 0, 5 \rangle$??? 15

1.

a

f

b

e

c

d

Braille letter

Monotonicity and extrema, group *Omega* ω -ii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{6} + \frac{6}{x}$, $x_0 = 1$??? increasing

(b) $g(x) = x^2 e^{-6x+4}$??? $(-\infty, 0)$, $\langle \frac{1}{3}, \infty \rangle$

(c) $h(t) = \frac{\ln t}{8t^4}$??? 2

(d) $v(x) = -3 - \sqrt[5]{(-x+2)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 5x + 8}{x - 1}$??? 8

(f) $\omega(x) = -\frac{1}{3}x^3 - 3x^2 - 8x + 4$, $x \in \langle -7, -1 \rangle$??? 28

2.

a

f

b

e

c

d

Braille letter

Monotonicity and extrema, group *Omega* ω -iii

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{10} + \frac{10}{x}$, $x_0 = 8$??? decreasing

(b) $g(x) = x^2 e^{-8x-2}$??? $\langle \frac{1}{4}, 0 \rangle$

(c) $h(t) = \frac{\ln t}{4t^4}$??? 1

(d) $v(x) = -7 - \sqrt[3]{(3x-4)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 + 3x + 1}{x + 3}$??? -4

(f) $\omega(x) = -\frac{4}{3}x^3 + \frac{25}{2}x^2 - 25x + 6$, $x \in \langle -1, 6 \rangle$??? $-\frac{5}{4}$

3.

a

f

b

e

c

d

Braille letter

Monotonicity and extrema, group *Omega* ω -iv

- (a) Determine the monotonicity of the function $f(x)$ in the neighborhood of the point x_0 .

(b) Determine the intervals on which the function $g(x)$ is decreasing.

(c) Determine the number of stationary points of the function $h(t)$.

(d) Does the function $v(x)$ attain a local extremum?

(e) Determine the difference between the values of the local maximum and the local minimum of the function $\varphi(x)$.

(f) Determine the product of the x -coordinates of the points of the global maximum and global minimum of the function $\omega(x)$ on the given interval.

- (a) $f(x) = \frac{x}{8} + \frac{8}{x}$, $x_0 = -12$.. ??? .. increasing

(b) $g(x) = x^2 e^{-2x+7}$.. ??? .. $(-\infty, 0)$, $\langle 1, \infty \rangle$

(c) $h(t) = \frac{\ln t}{2t^5}$??? 2

(d) $v(x) = 4 - \sqrt[3]{(-4x-3)^2}$??? nie

(e) $\varphi(x) = \frac{x^2 - 3x - 69}{x + 7}$??? -4

(f) $\omega(x) = x^3 - 2x^2 - 4x - 4$, $x \in \langle -1, 4 \rangle$??? 8

4.

a

f

b

e

c

d

Braille letter

Monotonicity and extrema (solutions)

α	$i : \mathbf{H}$	(a) increasing✓	(b) $(-\infty, 0), \langle 1/2, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-20\checkmark$	(f) 15✓
	$ii : \mathbf{O}$	(a) increasing✓	(b) $\langle -1/2, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-16\checkmark$	(f) 24✓
	$iii : \mathbf{P}$	(a) increasing✓	(b) $(-\infty, 0), \langle 2/9, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-16\checkmark$	(f) 8✓
	$iv : \mathbf{E}$	(a) increasing✓	(b) $(-\infty, 0), \langle 2/7, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-8\checkmark$	(f) 15✓
β	$i : \mathbf{F}$	(a) increasing✓	(b) $(-\infty, 0), \langle 2/3, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) 4✓
	$ii : \mathbf{E}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) 15✓
	$iii : \mathbf{E}$	(a) decreasing✓	(b) $\langle -2, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) $-1\checkmark$
	$iv : \mathbf{T}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1/3, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-16\checkmark$	(f) 0✓
γ	$i : \mathbf{S}$	(a) decreasing✓	(b) $\langle -2/3, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-8\checkmark$	(f) $12/5\checkmark$
	$ii : \mathbf{O}$	(a) increasing✓	(b) $\langle -2, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-8\checkmark$	(f) 24✓
	$iii : \mathbf{U}$	(a) increasing✓	(b) $(-\infty, 0), \langle 2/7, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-8\checkmark$	(f) 3✓
	$iv : \mathbf{L}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1/2, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) 3✓
δ	$i : \mathbf{H}$	(a) decreasing✓	(b) $\langle -2/3, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-20\checkmark$	(f) $-1/4\checkmark$
	$ii : \mathbf{A}$	(a) increasing✓	(b) $\langle -2/5, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-20\checkmark$	(f) $-2\checkmark$
	$iii : \mathbf{L}$	(a) decreasing✓	(b) $\langle -2/3, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-8\checkmark$	(f) $-2/3\checkmark$
	$iv : \mathbf{F}$	(a) increasing✓	(b) $\langle -2/5, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-12\checkmark$	(f) 24✓
ϵ	$i : \mathbf{R}$	(a) increasing✓	(b) $\langle -1/3, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) 6✓
	$ii : \mathbf{A}$	(a) decreasing✓	(b) $\langle -1/2, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) 28✓
	$iii : \mathbf{I}$	(a) increasing✓	(b) $\langle -2/5, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) 4✓
	$iv : \mathbf{N}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-12\checkmark$	(f) 8✓
ζ	$i : \mathbf{B}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 2/9, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-20\checkmark$	(f) 8✓
	$ii : \mathbf{I}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 2/5, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-8\checkmark$	(f) $-5/4\checkmark$
	$iii : \mathbf{R}$	(a) increasing✓	(b) $(-\infty, 0), \langle 2/3, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) 4✓
	$iv : \mathbf{D}$	(a) decreasing✓	(b) $\langle -2, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-12\checkmark$	(f) 0✓
η	$i : \mathbf{R}$	(a) increasing✓	(b) $\langle -2/7, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-16\checkmark$	(f) 15✓
	$ii : \mathbf{O}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 2/3, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) 8✓
	$iii : \mathbf{A}$	(a) decreasing✓	(b) $\langle -1/2, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-16\checkmark$	(f) 10✓
	$iv : \mathbf{D}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 2/5, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-16\checkmark$	(f) 8✓
θ	$i : \mathbf{S}$	(a) increasing✓	(b) $\langle -2/5, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-24\checkmark$	(f) $-1\checkmark$
	$ii : \mathbf{A}$	(a) increasing✓	(b) $\langle -1, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-8\checkmark$	(f) 2✓
	$iii : \mathbf{V}$	(a) increasing✓	(b) $(-\infty, 0), \langle 2, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) 4✓
	$iv : \mathbf{E}$	(a) increasing✓	(b) $(-\infty, 0), \langle 2/5, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) 8✓
ι	$i : \mathbf{D}$	(a) decreasing✓	(b) $\langle -1, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-12\checkmark$	(f) 24✓
	$ii : \mathbf{O}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-24\checkmark$	(f) 70✓
	$iii : \mathbf{O}$	(a) increasing✓	(b) $(-\infty, 0), \langle 1/3, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-12\checkmark$	(f) 48✓
	$iv : \mathbf{R}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-8\checkmark$	(f) $-1/3\checkmark$
κ	$i : \mathbf{C}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 2, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-16\checkmark$	(f) 8✓
	$ii : \mathbf{L}$	(a) increasing✓	(b) $\langle -2, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-12\checkmark$	(f) $9/4\checkmark$
	$iii : \mathbf{I}$	(a) decreasing✓	(b) $\langle -1/2, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-24\checkmark$	(f) 70✓
	$iv : \mathbf{P}$	(a) decreasing✓	(b) $\langle -1/4, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) $-1\checkmark$
λ	$i : \mathbf{L}$	(a) increasing✓	(b) $(-\infty, 0), \langle 1, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-8\checkmark$	(f) 0✓
	$ii : \mathbf{O}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1/2, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-20\checkmark$	(f) 8✓
	$iii : \mathbf{V}$	(a) increasing✓	(b) $(-\infty, 0), \langle 1/2, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-24\checkmark$	(f) 7✓
	$iv : \mathbf{E}$	(a) increasing✓	(b) $(-\infty, 0), \langle 2/9, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-12\checkmark$	(f) $-2/7\checkmark$
μ	$i : \mathbf{D}$	(a) decreasing✓	(b) $\langle -1/4, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-16\checkmark$	(f) 28✓
	$ii : \mathbf{E}$	(a) decreasing✓	(b) $\langle -1/3, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-4\checkmark$	(f) 21✓
	$iii : \mathbf{S}$	(a) decreasing✓	(b) $\langle -2, 0 \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-12\checkmark$	(f) 3✓
	$iv : \mathbf{K}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1, \infty \rangle$ ✓	(c) 1✓	(d) áno✓	(e) $-16\checkmark$	(f) 24✓

Monotonicity and extrema (solutions)

ν	$i : \mathbf{G}$	(a) increasing✓	(b) $(-\infty, 0), \langle 1/4, \infty \rangle$ ✓	(c) 1 $\times$	(d) áno $\times$	(e) $-12\checkmark$	(f) 24 $\checkmark$
	$ii : \mathbf{O}$	(a) decreasing✓	(b) $\langle -2/5, 0 \rangle$ $\times$	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-8\checkmark$	(f) $-1\mathbf{x}$
	$iii : \mathbf{L}$	(a) increasing✓	(b) $\langle -2/5, 0 \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-8\mathbf{x}$	(f) $5/2\mathbf{x}$
	$iv : \mathbf{D}$	(a) increasing✓	(b) $\langle -1/2, 0 \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-12\checkmark$	(f) 24 $\checkmark$
ξ	$i : \mathbf{L}$	(a) decreasing✓	(b) $\langle -2/7, 0 \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-8\mathbf{x}$	(f) 40 $\mathbf{x}$
	$ii : \mathbf{E}$	(a) decreasing✓	(b) $\langle -1/2, 0 \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-16\checkmark$	(f) 3 $\mathbf{x}$
	$iii : \mathbf{A}$	(a) increasing✓	(b) $\langle -1/2, 0 \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-8\mathbf{x}$	(f) $-1/3\mathbf{x}$
	$iv : \mathbf{D}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1/3, \infty \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-8\checkmark$	(f) 24 $\checkmark$
o	$i : \mathbf{S}$	(a) decreasing $\times$	(b) $(-\infty, 0), \langle 1/2, \infty \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-16\mathbf{x}$	(f) 3 $\checkmark$
	$ii : \mathbf{H}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1, \infty \rangle$ ✓	(c) 1 $\times$	(d) áno $\times$	(e) $-12\checkmark$	(f) 24 $\mathbf{x}$
	$iii : \mathbf{O}$	(a) increasing✓	(b) $(-\infty, 0), \langle 1/2, \infty \rangle$ $\times$	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-4\checkmark$	(f) 3 $\mathbf{x}$
	$iv : \mathbf{W}$	(a) increasing $\times$	(b) $\langle -1, 0 \rangle$ ✓	(c) 1 $\times$	(d) áno✓	(e) $-20\checkmark$	(f) $-2\checkmark$
π	$i : \mathbf{R}$	(a) decreasing✓	(b) $\langle -2/3, 0 \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-4\checkmark$	(f) 8 $\mathbf{x}$
	$ii : \mathbf{U}$	(a) increasing✓	(b) $(-\infty, 0), \langle 2/7, \infty \rangle$ $\times$	(c) 1 $\checkmark$	(d) áno✓	(e) $-8\mathbf{x}$	(f) 48 $\mathbf{x}$
	$iii : \mathbf{L}$	(a) increasing✓	(b) $\langle -1/4, 0 \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-8\mathbf{x}$	(f) $-2\mathbf{x}$
	$iv : \mathbf{E}$	(a) decreasing✓	(b) $\langle -2/7, 0 \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-16\checkmark$	(f) 15 $\mathbf{x}$
ρ	$i : \mathbf{T}$	(a) decreasing $\times$	(b) $(-\infty, 0), \langle 2/5, \infty \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-4\checkmark$	(f) 24 $\checkmark$
	$ii : \mathbf{R}$	(a) decreasing✓	(b) $\langle -1/4, 0 \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-24\checkmark$	(f) $-1\mathbf{x}$
	$iii : \mathbf{I}$	(a) decreasing $\times$	(b) $(-\infty, 0), \langle 2/5, \infty \rangle$ ✓	(c) 1 $\times$	(d) áno $\times$	(e) $-4\checkmark$	(f) 70 $\checkmark$
	$iv : \mathbf{P}$	(a) increasing✓	(b) $(-\infty, 0), \langle 1/2, \infty \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-8\mathbf{x}$	(f) 15 $\checkmark$
σ	$i : \mathbf{S}$	(a) increasing $\times$	(b) $\langle -2/5, 0 \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-12\mathbf{x}$	(f) $-2\checkmark$
	$ii : \mathbf{U}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1, \infty \rangle$ $\times$	(c) 1 $\checkmark$	(d) áno✓	(e) $-12\mathbf{x}$	(f) $-1\mathbf{x}$
	$iii : \mathbf{I}$	(a) increasing $\times$	(b) $\langle -2/3, 0 \rangle$ ✓	(c) 1 $\times$	(d) áno $\times$	(e) $-4\mathbf{x}$	(f) 8 $\checkmark$
	$iv : \mathbf{T}$	(a) decreasing $\times$	(b) $\langle -2/5, 0 \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-8\checkmark$	(f) $3/2\checkmark$
τ	$i : \mathbf{N}$	(a) increasing✓	(b) $(-\infty, 0), \langle 1, \infty \rangle$ $\times$	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-4\checkmark$	(f) $-1/2\checkmark$
	$ii : \mathbf{O}$	(a) decreasing✓	(b) $\langle -1/2, 0 \rangle$ $\times$	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-12\checkmark$	(f) 54 $\mathbf{x}$
	$iii : \mathbf{T}$	(a) decreasing $\times$	(b) $\langle -2, 0 \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-4\checkmark$	(f) 88 $\checkmark$
	$iv : \mathbf{E}$	(a) increasing✓	(b) $(-\infty, 0), \langle 2/3, \infty \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-4\checkmark$	(f) $25/2\mathbf{x}$
v	$i : \mathbf{Y}$	(a) decreasing✓	(b) $\langle -2/3, 0 \rangle$ $\times$	(c) 1 $\checkmark$	(d) áno✓	(e) $-20\checkmark$	(f) 2 $\checkmark$
	$ii : \mathbf{E}$	(a) increasing✓	(b) $(-\infty, 0), \langle 2, \infty \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-4\checkmark$	(f) 24 $\mathbf{x}$
	$iii : \mathbf{A}$	(a) increasing✓	(b) $\langle -1/3, 0 \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-12\mathbf{x}$	(f) 7 $\mathbf{x}$
	$iv : \mathbf{R}$	(a) increasing✓	(b) $(-\infty, 0), \langle 1/2, \infty \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-12\checkmark$	(f) $25/2\mathbf{x}$
ϕ	$i : \mathbf{G}$	(a) increasing✓	(b) $\langle -2/3, 0 \rangle$ ✓	(c) 1 $\times$	(d) áno $\times$	(e) $-8\checkmark$	(f) 7 $\checkmark$
	$ii : \mathbf{A}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1/4, \infty \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-20\mathbf{x}$	(f) $-2\mathbf{x}$
	$iii : \mathbf{M}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1, \infty \rangle$ $\times$	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-12\mathbf{x}$	(f) 2 $\checkmark$
	$iv : \mathbf{E}$	(a) increasing✓	(b) $(-\infty, 0), \langle 2, \infty \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-8\checkmark$	(f) $20/3\mathbf{x}$
χ	$i : \mathbf{T}$	(a) increasing $\times$	(b) $(-\infty, 0), \langle 2, \infty \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-8\checkmark$	(f) $20/3\checkmark$
	$ii : \mathbf{R}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1, \infty \rangle$ ✓	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-20\checkmark$	(f) 24 $\mathbf{x}$
	$iii : \mathbf{E}$	(a) decreasing✓	(b) $\langle -2/5, 0 \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-8\checkmark$	(f) 24 $\mathbf{x}$
	$iv : \mathbf{E}$	(a) increasing✓	(b) $\langle -1, 0 \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-4\checkmark$	(f) $3/2\mathbf{x}$
ψ	$i : \mathbf{N}$	(a) decreasing✓	(b) $\langle -2/5, 0 \rangle$ $\times$	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-4\checkmark$	(f) 24 $\checkmark$
	$ii : \mathbf{I}$	(a) increasing $\times$	(b) $(-\infty, 0), \langle 1, \infty \rangle$ ✓	(c) 1 $\times$	(d) áno $\times$	(e) $-16\mathbf{x}$	(f) 3 $\checkmark$
	$iii : \mathbf{N}$	(a) decreasing✓	(b) $\langle -1/4, 0 \rangle$ $\times$	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-12\checkmark$	(f) 54 $\checkmark$
	$iv : \mathbf{E}$	(a) increasing✓	(b) $\langle -2/7, 0 \rangle$ $\times$	(c) 1 $\times$	(d) áno $\times$	(e) $-4\checkmark$	(f) 3 $\mathbf{x}$
ω	$i : \mathbf{W}$	(a) increasing $\times$	(b) $\langle -2, 0 \rangle$ ✓	(c) 1 $\times$	(d) áno✓	(e) $-4\checkmark$	(f) 15 $\checkmark$
	$ii : \mathbf{I}$	(a) decreasing $\times$	(b) $(-\infty, 0), \langle 1/3, \infty \rangle$ ✓	(c) 1 $\times$	(d) áno $\times$	(e) $-8\mathbf{x}$	(f) 28 $\checkmark$
	$iii : \mathbf{N}$	(a) decreasing✓	(b) $(-\infty, 0), \langle 1/4, \infty \rangle$ $\times$	(c) 1 $\checkmark$	(d) áno $\times$	(e) $-4\checkmark$	(f) $-5/4\checkmark$
	$iv : \mathbf{G}$	(a) increasing✓	(b) $(-\infty, 0), \langle 1, \infty \rangle$ ✓	(c) 1 $\times$	(d) áno $\times$	(e) $-4\checkmark$	(f) 8 $\checkmark$